

RentalPro

Digital Rental Management System

DELIVERABLE 5

Feature Completion Report

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Sections	Introduction · Gap Analysis · Implementation · Design and Data Changes · Testing · Requirements Traceability · Conclusion
Appendix	Appendix A — Verifying the Deliverable 5 Features
Prior Deliverables	Deliverable 1 — SRS · Deliverable 2 — UML & Database Design · Deliverable 3 — Partially-Developed Application · Deliverable 4 — Complete Project Report

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1. Introduction

1.1 Purpose of Deliverable 5

Deliverable 3 delivered a working RentalPro application in which every one of the 25 use cases from the Software Requirements Specification (SRS) had a route, a screen and at least one automated test. A use case being reachable, however, is not the same as every functional requirement behind it being met. Deliverable 5 closed that remaining distance: it audited the running application against all 44 functional requirements (FR-001 to FR-044) and the 25 use cases, identified what was missing or only partly built, and implemented, tested and documented each gap.

This report records that work: what the gaps were (Section 2), how each one was closed (Section 3), what it changed in the design and data model (Section 4), how it was tested (Section 5), and where every functional requirement now stands (Section 6).

1.2 Starting Point

At the start of Deliverable 5 the application already covered authentication and role-based access, property and unit creation, tenant onboarding and invitations, lease upload, rent recording and statements, online payment in dev mode, the maintenance lifecycle, the four background jobs and portfolio reporting, verified by 93 automated tests. The audit found eleven places where the SRS asked for more than the application did — for example, units could be created but not edited afterwards, no photos could be attached anywhere, and there was no way to blacklist a tenant, configure a late fee, or manage users as an administrator.

1.3 Approach

- **Audit first.** Each FR and use case was read against the routes, templates and tests of the running application, and every shortfall was written down with the requirement it failed.
- **One vertical slice per gap.** Each gap was built end to end — model fields, route, template, access control, audit logging and tests — and checked in the browser before moving to the next.
- **Additive changes only.** Every change added fields with safe defaults or new routes; nothing existing was renamed or removed, so the 13-table schema and all 93 existing tests stayed valid.
- **Re-verify the documents.** Once the code was complete, every figure in this report was produced by running it. The earlier reports keep the figures of the build they were submitted with: Deliverable 3 stands exactly as submitted, recording 93 tests at 77% coverage, and Deliverable 4 records the same figures, with a closing appendix on what changed afterwards.

1.4 Results at a Glance

Measure	Before Deliverable 5	After Deliverable 5
Gaps against the SRS	11 identified	11 closed
Use cases with a route, screen and test	25 / 25	25 / 25
Database tables	13	13 (new fields only, no new tables)
Automated tests	93	124 (31 new, in tests/test_deliverable5.py)

Measure	Before Deliverable 5	After Deliverable 5
Statement coverage	77%	79%
Test failures	0	0

Note on later work: since Deliverable 5 the system has grown further — a billing engine, a property map, a redesigned Harare interface, Harare demonstration data, and English, Shona and Arabic interface languages (see Deliverable 4, Appendix C, and the Deliverable 6 report). Two requirements this report records as met after Deliverable 5 — a tenant history per unit (FR-015) and a maintenance summary per period (FR-036) — were completed in September 2026 (Section 7.3). Today the suite has 298 tests at 85% coverage, and all 31 Deliverable 5 tests still pass. Screenshots in this report were taken from the current application, so they show the newer interface around the Deliverable 5 features. Since then, Arabic has been extended to every screen, message, email, SMS and PDF, every amount shows its currency, and the system has been hardened for production with security headers, a required secret key, Alembic database migrations and optional S3 file storage (see the Deliverable 6 final report, Section 3.6).

2. Gap Analysis

2.1 How the Gaps Were Found

Every functional requirement was checked in three ways: is there a route or scheduled job that performs it; does a user of the right role reach it from the interface; and does an automated test prove it. A requirement failing any of the three was recorded as a gap. The use cases were then re-read the same way, which added the lease-document download (UC-10) and CSV export (UC-17) — both named in a use case but not in a single FR on their own.

2.2 The Eleven Gaps

#	Gap	Requirement	Before Deliverable 5
1	Edit an existing unit's details and status	FR-005, FR-004, UC-04	Units could be created but never edited
2	Property photos (up to 10) and unit photos (up to 5)	FR-003	No photo storage anywhere
3	Maintenance photos on submission and on completion (up to 3 each)	FR-028, UC-18, UC-20	Text-only requests
4	Tenant ID document upload	UC-07	Not captured
5	Lease document download for management and the tenant	UC-10, FR-012	Uploaded PDFs could not be retrieved
6	Blacklist / unblacklist a tenant, blocking future leases	FR-016	Not available
7	Administrator user and role management	UC-02	Roles could only be set in the database
8	Staff account creation with a welcome email	FR-044	Only tenant accounts could be created
9	Configurable late fees per property (fixed or percentage)	FR-022	No late fees
10	CSV export of a lease's rent history	UC-17	On-screen history only
11	Excel export of the portfolio report	FR-043, UC-25	PDF export only

3. Implementation

Each subsection below states the requirement, what was built, the rules the system now enforces, and where it lives in the code.

3.1 Editing Units and Occupancy Rules (FR-004, FR-005)

A new Edit Unit form (/properties/<id>/units/<id>/edit) lets an Owner, Manager or Administrator change a unit's floor, type, size, monthly rent and deposit at any time, add photos, and set its status. Two rules keep the occupancy figures honest:

- A unit cannot be set to Occupied by hand — it becomes Occupied only when a lease is activated against it.
- A unit with an active lease cannot be set back to Vacant; the status field is locked and explains why.
- Under Maintenance can be set freely on a vacant unit, so a unit being repaired is not offered to new tenants.

Figure 1: Edit Unit — the status is locked while a lease is active

3.2 Property and Unit Photos (FR-003)

The Add/Edit Property form accepts up to 10 photos per property, and the Add/Edit Unit forms up to 5 per unit, matching the SRS limits exactly. The limit counts photos already stored, so a property with 8 photos can take only 2 more. Each file must be a genuine JPEG or PNG of at most 5 MB: the system reads the file's first bytes and compares them with the JPEG (FF D8 FF) and PNG (89 50 4E 47) signatures, so a renamed PDF or executable is rejected whatever its extension. Rejected files are reported by name rather than silently dropped.

Figure 2: Edit Property — late-fee settings and the photo upload (up to 10, JPG/PNG, 5 MB each)

3.3 Maintenance Photos (FR-028)

A tenant can attach up to 3 photos when submitting a maintenance request, so staff can see the problem before they arrive, and the assigned staff member can attach up to 3 completion photos when marking the work done. Both sets appear on the ticket for the tenant and for management, and are served only to users allowed to see that ticket.

3.4 Tenant ID Document and Lease Download (UC-07, UC-10, FR-012)

The Create Tenant Profile form now accepts an identification document (JPEG, PNG or PDF, up to 10 MB). Signed lease PDFs, which could already be uploaded, can now be downloaded again — by management, or by the tenant the lease belongs to. Both documents are served through authenticated routes that check the requester's role and, for a tenant, that the document is their own (`assert_tenant_self`), so one tenant can never open another's ID or lease.

3.5 Tenant Blacklisting (FR-016)

An Owner, Manager or Administrator can flag a tenant as blacklisted from the tenant's profile. A reason is required and is shown wherever the flag appears. While the flag is set, the tenant cannot be given a new lease — the lease-upload form refuses with the reason — but a lease already in force is left alone, as the SRS asks only for future leases to be prevented. Removing the flag is one click, and both actions are written to the audit log.

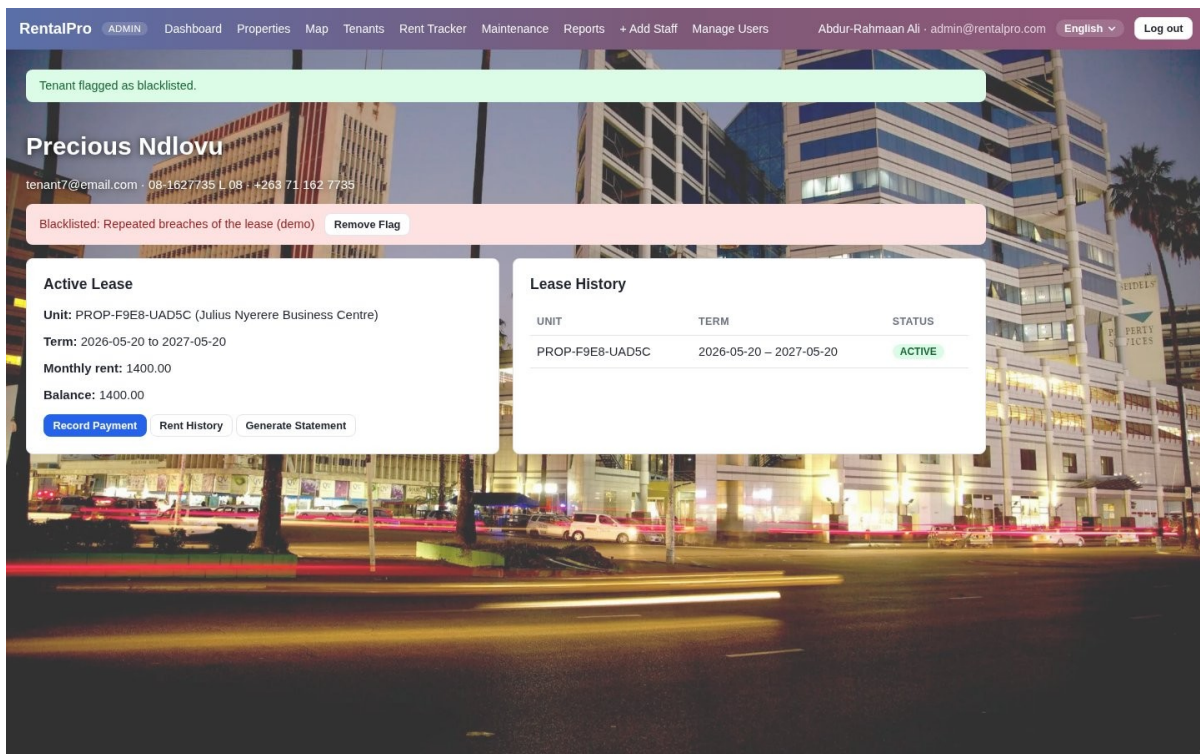


Figure 3: A blacklisted tenant — the reason is shown with a Remove Flag action; the current lease is unaffected

3.6 Administrator User and Role Management (UC-02)

The Manage Users & Roles console (/admin/users, Administrator only) lists every account with its role and status, and lets the Administrator change a role or deactivate an account; a deactivated account cannot log in. One safeguard protects the portfolio: the system will not demote or deactivate the last active Property Owner, since nobody would then be able to add properties. Every change is audited with the old and new values, and the affected user is emailed.

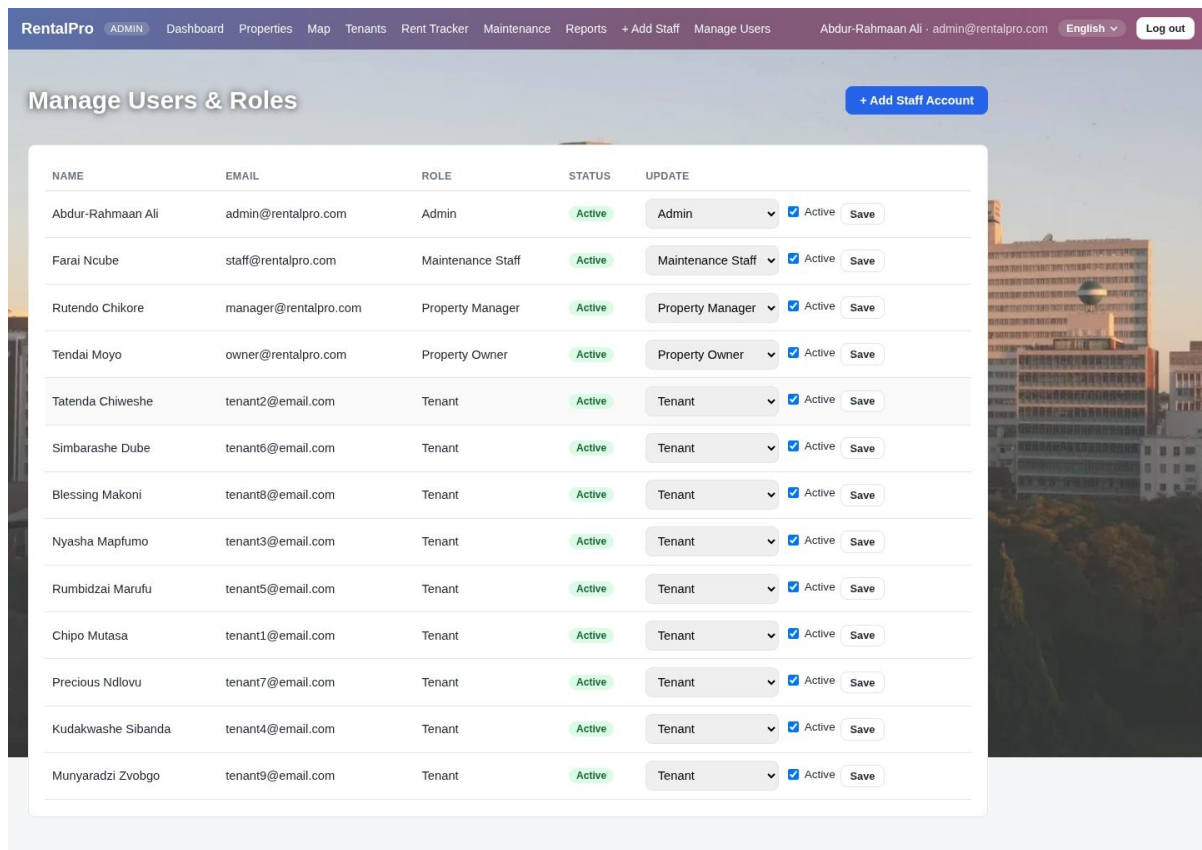


Figure 4: Manage Users & Roles — role and active status for every account

3.7 Staff Account Creation (FR-044)

Owners, Managers and Administrators can create a Maintenance Staff account from “+ Add Staff”. The new user receives a welcome email with a link to set their own password, valid for 24 hours; an Administrator can later change the account’s role in the console if it should be a Manager instead. The SRS asks for a welcome email “with login credentials”; RentalPro deliberately sends a set-password link instead, so no password is ever sent by email.

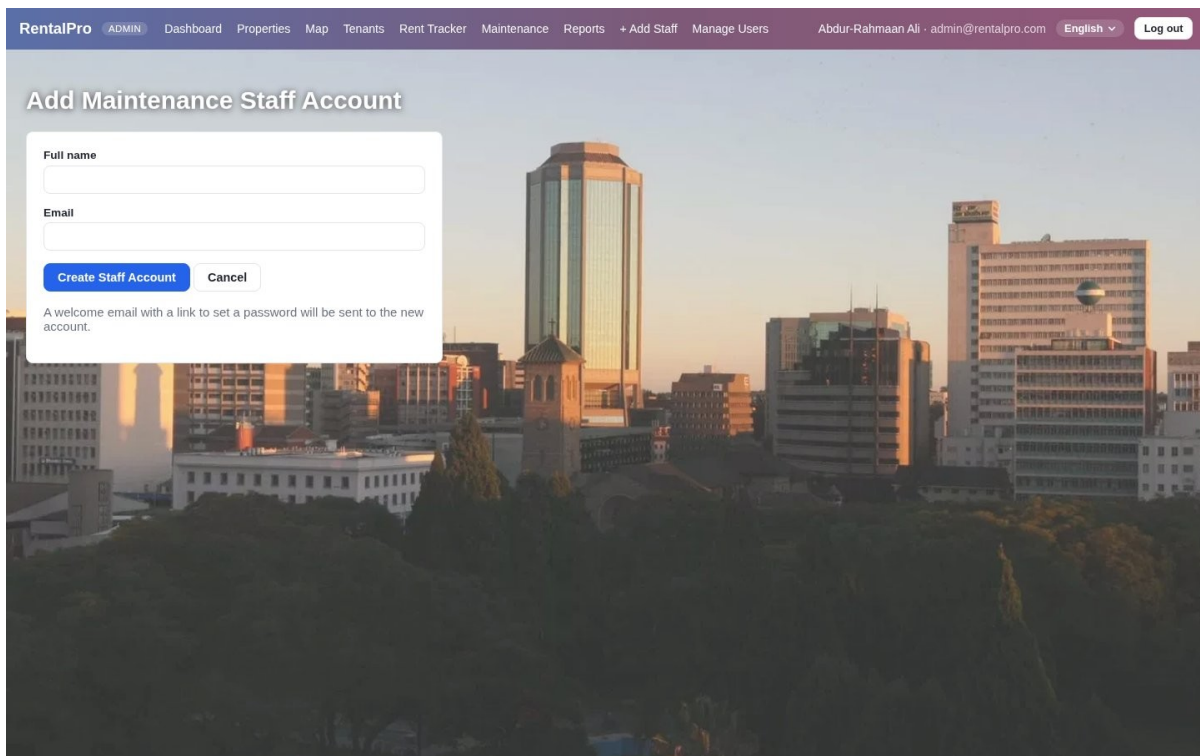


Figure 5: Add Maintenance Staff Account

3.8 Configurable Late Fees (FR-022)

Each property now has a late-fee type — None, Fixed or Percentage — and an amount (Figure 1). The fee is calculated, never stored, by `Lease.late_fee_due()`:

- It is zero unless the current billing period is past its due date and the lease is not fully paid for the periods due so far.
- A Fixed fee adds the configured amount; a Percentage fee adds that percentage of the lease's current monthly rent.
- The fee is folded into the lease balance, so it shows on the rent tracker, statements and the tenant portal, and disappears once the overdue amount is paid.
- New properties default to None, so no existing lease changed when the feature was introduced.

Example: a \$850 lease on a property with a 5% late fee, unpaid past its due date, carries a \$42.50 fee ($850 \times 5\%$); with a Fixed fee of \$25 it carries \$25.

3.9 Rent History CSV Export (UC-17)

Every lease's rent history page has an Export CSV button that downloads each payment's date, type, amount, method, receipt number and notes — ready for a spreadsheet or accounting package. Management can export any lease they can see; a tenant can export only their own.

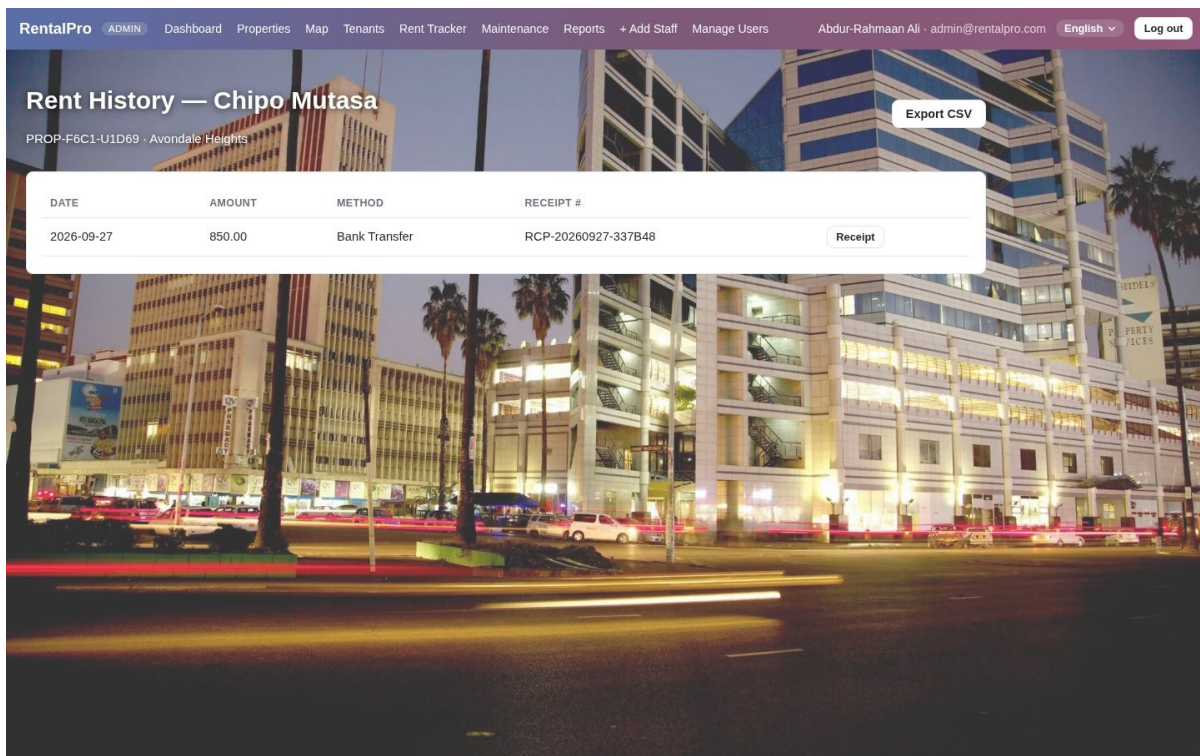


Figure 6: Rent history with the Export CSV button

3.10 Excel Report Export (FR-043)

The Property Performance Report, which could already be exported as PDF, can now also be exported as an Excel workbook, for the whole portfolio or a single property. The workbook has two sheets — Portfolio Summary and By Property — built with openpyxl, so an owner can sort, filter and chart the figures themselves.

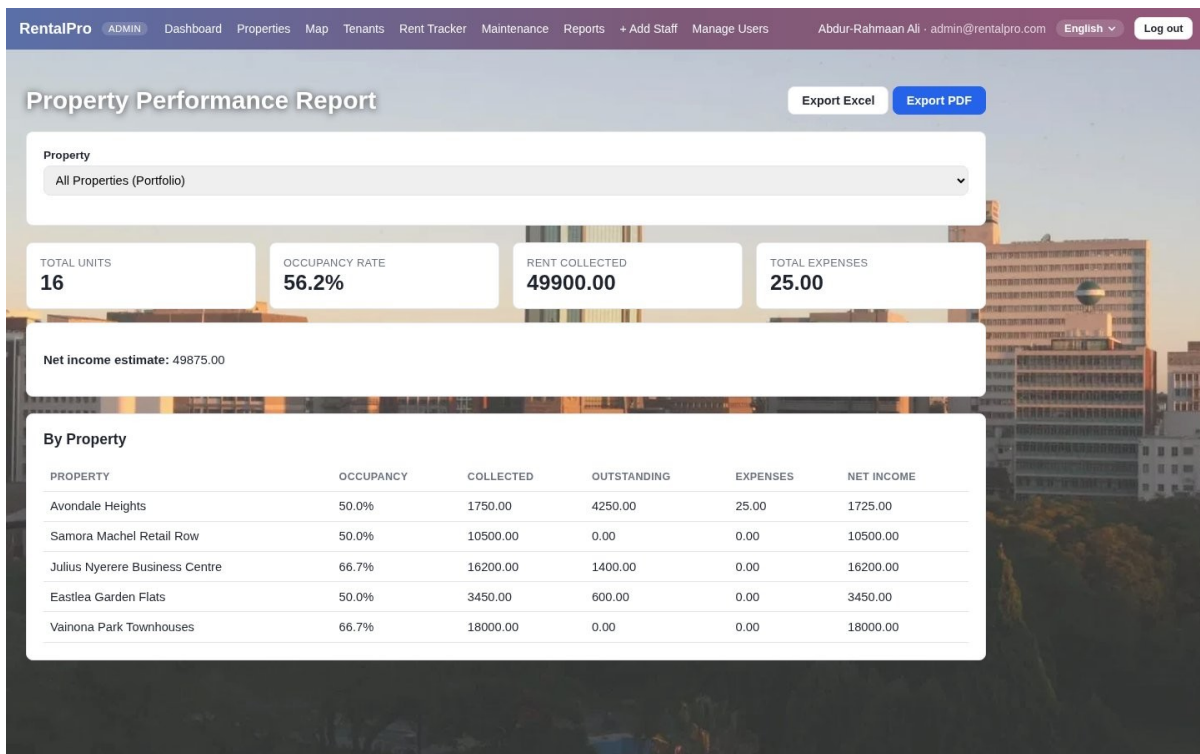


Figure 7: Property Performance Report with Export Excel and Export PDF

4. Design and Data Changes

4.1 Data Model

All eleven gaps were closed without a new table. Photos and documents are stored as file paths on the record they belong to, which kept the 13-table schema from Deliverable 2 intact:

Model	New field(s)	Type and default	Used by
Property	photo_paths	JSON list, default []	Property photos (3.2)
Property	late_fee_type, late_fee_amount	NONE / FIXED / PERCENTAGE; amount 0	Late fees (3.8)
Unit	photo_paths	JSON list, default []	Unit photos (3.2)
Tenant	is_blacklisted, blacklist_reason	Boolean false; text up to 300 characters	Blacklisting (3.5)
Tenant	id_document_path	Text, nullable	ID document (3.4)
MaintenanceRequest	photo_paths, completion_photo_paths	JSON lists, default []	Maintenance photos (3.3)
Lease	late_fee_due() (method, not a column)	Calculated	Balance with late fee (3.8)

4.2 One Validator and One Storage Gateway

All five upload points share the same two pieces of code rather than each having its own:

- **validate_upload() in security.py** checks the extension, the size limit for that field and the file's magic bytes. It already existed for lease PDFs; Deliverable 5 extended it to JPEG and PNG.
- **services/storage.py** provides save_upload(), save_uploads() and serve_upload(). Files are saved under the uploads folder with generated names, and served only through routes that have already checked the user's role and ownership. Moving to cloud storage later means changing this one module, not the routes.

Upload	Accepted types	Size limit	Count limit
Property photos	JPG, PNG	5 MB each	10 per property
Unit photos	JPG, PNG	5 MB each	5 per unit
Maintenance photos	JPG, PNG	5 MB each	3 on submission, 3 on completion
Tenant ID document	JPG, PNG, PDF	10 MB	1
Signed lease	PDF	20 MB	1 per lease

4.3 Access Control on the New Routes

Route	Roles allowed	Further check
/properties/<id>/edit, /properties/<id>/units/<id>/edit	Admin, Owner, Manager	assert_owner — an Owner edits only their own properties

Route	Roles allowed	Further check
/tenants/<id>/blacklist, /unblacklist	Admin, Owner, Manager	Reason required to blacklist
/tenants/<id>/id-document, /lease-document/<lease_id>	Management and Tenant	assert_tenant_self — a tenant sees only their own
/admin/users, /admin/users/<id>/edit	Admin only	Last active Owner cannot be removed
/admin/staff/add	Admin, Owner, Manager	Duplicate email rejected
/rent/<lease_id>/history.csv	Management and Tenant	assert_tenant_self
/reports/export.xlsx	Admin, Owner, Manager	assert_owner when a single property is chosen

Documents are served through these authenticated routes rather than through time-limited signed URLs (NFR-012). With files on local disk this gives the same protection — no URL works without a logged-in, authorised session — and signed URLs remain the natural choice once storage moves to the cloud.

4.4 Audit Trail

Each new action writes to the append-only audit log with the acting user and time: unit_updated, tenant_blacklisted, tenant_unblacklisted, user_role_changed (with the old and new role and status), staff_created, rent_history_exported and report_exported.

5. Testing

5.1 The Deliverable 5 Test Module

All eleven gaps are covered by tests/test_deliverable5.py — 31 tests that drive real HTTP requests through Flask's test client (sessions, CSRF, role checks and the database), plus direct model tests for the late-fee arithmetic.

Area	Tests	What they prove
Unit and property editing	4	Fields update; Vacant is refused while a lease is active; Under Maintenance is allowed; late-fee settings save
Photo uploads	5	Property, unit and maintenance photos upload and are served; completion photos save; a non-image is rejected; PNG accepted
Blacklist and ID document	5	Blacklisting blocks a new lease; removing the flag allows it again; a reason is required; the ID document is downloadable by the right people and blocked for another tenant
User and role management	7	Admin lists users; non-admins are refused; the last active Owner cannot be removed; roles change when another Owner exists; a user can be deactivated; staff accounts are created; duplicate emails rejected
Late fees	4	Fixed and percentage fees apply once overdue; no fee before the due date; no fee when fully paid
CSV and Excel export	5	Management and the tenant can export the CSV; another tenant cannot; the Excel workbook has the right sheets, for the portfolio and for one property
Lease document download	1	Management can download a signed lease

5.2 Results

Run	Tests	Passed	Failed	Coverage
End of Deliverable 5	124	124	0	79%
Deliverable 5 module today	31	31	0	—
Full suite today (October 2026)	298	298	0	85%

The whole suite runs offline in about two minutes, with no payment, email or SMS account needed; the same tests run on every push in GitHub Actions.

6. Requirements Traceability

The table traces every functional requirement in the SRS to where it is implemented and a test that proves it. “D3” means it was already met at Deliverable 3; “D5” means Deliverable 5 completed it; “After D5” means it was completed later, in September 2026. Where the application meets a requirement in a different way, the table says so.

FR	Requirement	Status	Where	Evidence (test)
FR-001	Register a property	D3	/properties/add	add_property_generates_code
FR-002	Add units with details and rent	D3	/properties/<id>/add-unit	add_unit_generates_code_and_vacant_status
FR-003	Up to 10 property and 5 unit photos	D5	Property and unit forms	property_photo_upload_and_serve, unit_photo_upload_*
FR-004	Show Vacant / Occupied / Under Maintenance	D3 + D5	Unit status, Edit Unit	edit_unit_allows_under_maintenance
FR-005	Update unit details, rates and availability	D5	/properties/<id>/units/<id>/edit	edit_unit_updates_fields
FR-006	Unique property and unit codes	D3	Property / Unit models	property_code_format
FR-007	Occupancy dashboard	D3	/dashboard	property_occupancy_rate_with_units
FR-008	Archive units and properties	D3	Archive routes	archive_unit_blocked_by_active_lease
FR-009	Tenant profile with national ID and contacts	D3	/tenants/add	add_tenant_creates_user_and_tenant
FR-010	Self-registration by invitation link	D3	/tenants/invite	accept_invitation_creates_tenant_and_logs_in
FR-011	Link tenant to unit and lease	D3	/tenants/<id>/upload-lease	upload_lease_sets_unit_occupied
FR-012	Upload and secure storage of signed leases	D3 + D5	Upload; authorised download	upload_lease_rejects_non_pdf, lease_document_download_by_manager
FR-013	Lease dates, rent and deposit	D3	Lease model	upload_lease_sets_unit_occupied
FR-014	Lease expiry alerts at 60, 30 and 7 days	D3	job_lease_expiry_alerts	lease_expiry_alert_fires_at_30_days
FR-015	History of all tenants per unit	After D5	Unit History page (Sept 2026)	unit_history_lists_past_and_current_tenants_newest_first
FR-016	Blacklist a tenant with a reason	D5	/tenants/<id>/blacklist	blacklist_blocks_new_lease
FR-017	Tenant views profile, lease and payments	D3	/portal, /portal/history	lease_statement_api_allowed_for_own_tenant

FR	Requirement	Status	Where	Evidence (test)
FR-018	Record payments with method and receipt	D3	/rent/record	record_payment_generates_receipt_and_updates_balance
FR-019	Calculate outstanding balance	D3	Lease.balance	lease_balance_never_negative_when_overpaid
FR-020	Rent reminders 7 and 1 days before due	D3	job_rent_due_alerts	rent_due_alert_fires_at_7_and_1_day_thresholds
FR-021	Overdue notices on due date, then every 3 days	Variation	job_overdue_alerts: daily, escalating notices plus an owner summary	overdue_alert_creates_notification_and_owner_summary
FR-022	Late fees per property, fixed or %	D5	Property settings, Lease.late_fee_due()	late_fee_fixed_applies_once_overdue (+3)
FR-023	Monthly rent statement as PDF	D3	/rent/<id>/statement.pdf	test_account_statements.py
FR-024	Partial payments and remaining balance	D3	Lease.balance	lease_total_paid_accumulates
FR-025	Rent collection summary per property	D3	/reports	reports_page_and_json_export
FR-026	Online payment via gateway	D3	/portal/pay (Stripe, dev-mode fallback)	pay_rent_online_dev_mode_end_to_end
FR-027	Immutable audit of payments	D3	No edit/delete route; audit log	idempotent_payment_duplicate_gateway_ref_creates_one_payment
FR-028	Maintenance request with optional photos	D3 + D5	/maintenance/add	maintenance_request_photo_upload_and_serve
FR-029	Unique ticket number and timestamp	D3	MaintenanceRequest	ticket_number_format
FR-030	Assign with priority and target date	D3	/maintenance/<id>/update	maintenance_full_lifecycle
FR-031	Notify tenant on status change	D3	Email on assignment and each update	maintenance_full_lifecycle
FR-032	Six request stages	D3	MAINT_STATUSES	maintenance_full_lifecycle
FR-033	Staff progress notes and completion	D3 + D5	Update form; completion photos	maintenance_completion_photo_upload
FR-034	Tenant confirms or reopens	D3	close / reopen actions	maintenance_full_lifecycle (confirm)
FR-035	Log maintenance cost	D3	/maintenance/<id>/costs/add	maintenance_cost_accumulation
FR-036	Maintenance summary: open, in progress, closed per period	After D5	Reports → Maintenance Summary (Sept 2026)	maintenance_summary_groups_the_six_stages_into_three
FR-037	Escalate overdue requests	D3	job_escalate_overdue_maintenance	maintenance_escalation_fires_for_open_overdue

FR	Requirement	Status	Where	Evidence (test)
FR-038	Role-based access control	D3	role_required, assert_owner	test_auth.py (RBAC and IDOR tests)
FR-039	Dashboard per role	D3	/dashboard, /portal, work queue	login_success_redirects_by_role
FR-040	Arabic and English interfaces	After D5	English, Shona and Arabic, right to left (Sept 2026)	test_i18n.py
FR-041	Property performance report	D3	/reports	reports_page_and_json_export
FR-042	Audit trail of all actions	D3	AuditLog	audit_log_safe_outside_request_context
FR-043	Export reports as PDF or Excel	D3 + D5	/reports/export.pdf, .xlsx	reports_excel_export_structure
FR-044	Welcome email for new tenant or staff	D5 (variation)	Set-password link, not credentials	add_staff_account_by_manager

6.1 Summary

Status	Count	Requirements
Met in full	39	All others
Met in a different way	2	FR-021 (daily escalating overdue notices), FR-044 (set-password link instead of credentials)
Met after Deliverable 5	3	FR-015 (tenant history per unit), FR-036 (maintenance summary per period), FR-040 (interface languages) — all September 2026

All 25 use cases are implemented and tested. At the end of Deliverable 5, 41 of the 44 functional requirements were met (two of them in a different way by design), FR-015 and FR-036 were met only in part, and FR-040 was not yet built. All 44 are met today: FR-015 and FR-036 gained the screens they were missing, and FR-040 arrived with the interface languages (Section 7.3).

7. Conclusion

7.1 What Deliverable 5 Achieved

Deliverable 5 took RentalPro from “every use case works” to “every requirement has been checked”. Eleven gaps were closed — unit editing, photos, maintenance photos, ID documents, lease downloads, blacklisting, user management, staff accounts, late fees, CSV export and Excel export — each with its own tests, access control and audit logging, and without adding a single table or breaking an existing test. The suite grew from 93 to 124 tests at 79% coverage, while the Deliverable 3 and 4 reports keep the figures they were submitted with.

7.2 Remaining Work

- **Signed URLs for documents (NFR-012):** adopt once uploads move to cloud storage, which storage.py already isolates.
- **Exports of the maintenance summary:** the new summary (Section 7.3) is on screen only; adding it to the Excel and PDF exports would complete FR-043 for it too.

7.3 What Came Next

After Deliverable 5, the demonstration and presentation package (Deliverable 6) was produced, and the system was extended with a billing engine (charges, metering, pro rata and variable rent, automated billing, statements, receipts and expenses), a property map, a redesigned interface with Harare photography and demonstration data, and interface languages in English, Shona and Arabic — which also met FR-040. Those additions are described in Deliverable 4, Appendix C, and in the Deliverable 6 report.

In September 2026 the two requirements Deliverable 5 had left partly met were completed, with 13 new tests (tests/test_unit_history_and_maintenance_summary.py):

- **Tenant history per unit (FR-015):** a History button on every unit opens a list of each tenant who has occupied it — active, terminated and expired leases, newest first — with national ID, lease dates, rent and status (Figure 8). Owners see only their own properties; tenants and staff are refused.
- **Maintenance summary per period (FR-036):** the Reports page counts requests raised in a chosen date range as Open (submitted, acknowledged or assigned), In Progress or Closed (completed or closed), per property and in total, defaulting to the current month (Figure 9).

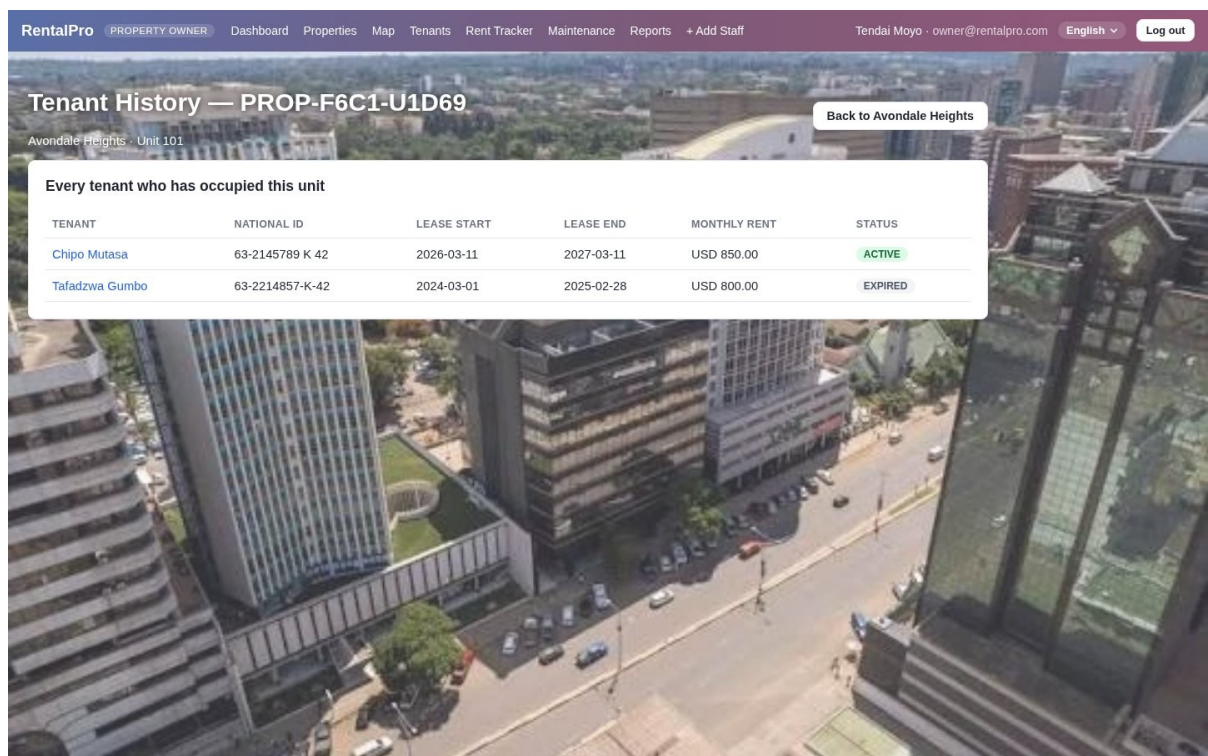


Figure 8: Tenant history for Avondale Heights, Unit 101 (FR-015)

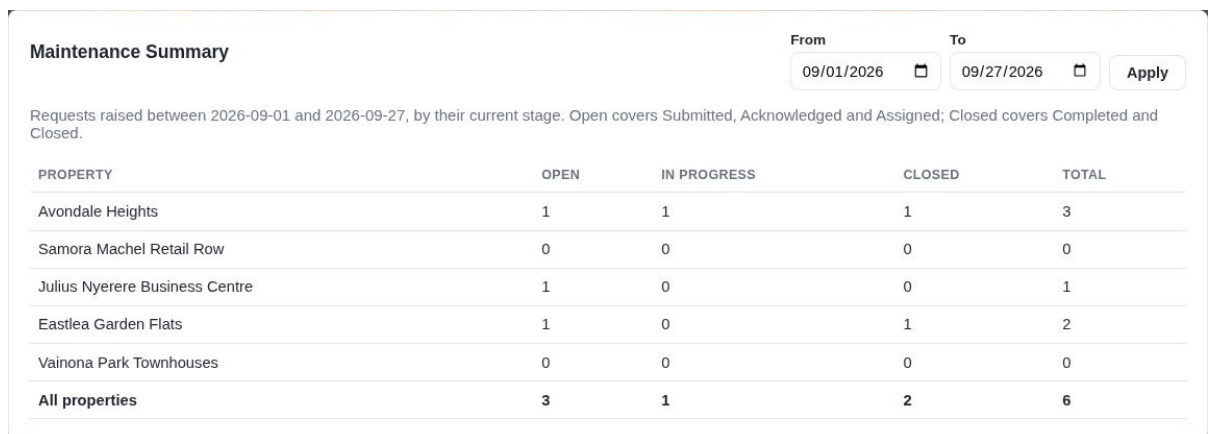


Figure 9: Maintenance Summary on the Reports page (FR-036)

Appendix A: Verifying the Deliverable 5 Features

A.1 Run the Tests

```
pytest tests/test_deliverable5.py -v          # the 31 Deliverable 5 tests
pytest tests/ --cov=backend                    # the whole suite with coverage
```

A.2 Try Them in the Application

Start the application (Deliverable 4, Appendix A) and sign in with the password demo123:

Feature	Sign in as	Where to go
Edit a unit, unit photos	owner@rentalpro.com	Properties → a property → Edit on a unit
Property photos, late fees	owner@rentalpro.com	Properties → a property → Edit Property
Blacklist a tenant	manager@rentalpro.com	Tenants → a tenant → enter a reason → Blacklist Tenant
Manage users and roles	admin@rentalpro.com	Manage Users
Create a staff account	admin@rentalpro.com	+ Add Staff
Maintenance photos	tenant1@email.com	Submit Request → attach photos
Rent history CSV	tenant1@email.com	Payment History → Export CSV
Excel report	owner@rentalpro.com	Reports → Export Excel
Tenant history per unit (after D5)	owner@rentalpro.com	Properties → a property → History on a unit
Maintenance summary (after D5)	owner@rentalpro.com	Reports → Maintenance Summary → choose dates → Apply